

Intermediate

1. A motorcyclist, travelling at a constant velocity of V m/s, passes a point A and sees roadworks ahead. He immediately applies the brakes and his subsequent velocity, v m/s, is given by $v = 24e^{\frac{-t}{6}}$, where t is the time in seconds after passing A . As he passes a point B his velocity has been halved.
- Find the time taken to travel from A to B . [3]
 - Find the acceleration of the motorcyclist as he passes B . [3]
 - Find the distance AB . [4]

[N21/II/8]

2. A motorcyclist, travelling along a straight road, passes a point A and, 10 seconds later, passes a point B . The motorcyclist's speed at B is twice his speed at A . For the journey from A to B , the motorcyclist's speed, v m/s, t seconds after passing A , is given by

$$v = 15 \left(\frac{t}{20} + e^{kt} \right), 0 \leq t \leq 10,$$

where k is a constant.

- Find the speed of the motorcyclist at B . [1]
- Find the distance between A and B . [6]
- Find the acceleration of the motorcyclist when $t = 2$. [3]

[N20/I/11]

3. A dot on a computer screen moves in a straight line so that, t seconds after leaving a fixed point O , its displacement, s cm, from O is modelled by $s = t^3 - 6t^2 + 9t$.
- Find the values of t at which the dot is instantaneously at rest. [3]
 - Find the acceleration of the dot when it first comes to instantaneous rest. [2]
 - Explain clearly why the total distance travelled by the dot in the interval $t = 0$ to $t = 4$ is **not** obtained by finding the value of s when $t = 4$. [2]
 - Find the total distance travelled by the dot in the interval $t = 0$ to $t = 4$. [3]

[N19/I/11]

4. A bungee jumper falls vertically from rest from a point O . Her velocity, v m/s, t seconds after leaving O , is such that $\frac{dv}{dt} = 10$. After 4 s, she reaches a point X . By integration, find
- her velocity at X , [2]
 - the distance OX . [2]

On reaching X , she then slows so that her velocity, V m/s, T seconds after reaching X , is such that $\frac{dV}{dT} = 10 - kT$, where k is a constant.

- Given that she first comes to rest at a point Y when $T = 3$, show that $k = \frac{140}{9}$. [3]

[N17/I/9]

5. A car, driven along a straight road, passes a signpost, A , with a speed of p km/h. A little later the car passes a second signpost, B , with a speed of 80 km/h. Between A and B the speed, v km/h, of the car is given by $v = 30e^{25t} + 20$, where t , the time after passing A , is measured in hours.
- State the value of p . [1]
 - Calculate, to the nearest second, the time taken to travel from A to B . [3]
 - Calculate the distance between A and B . [5]
 - Obtain an expression, in terms of t , for the acceleration of the car between A and B . [2]

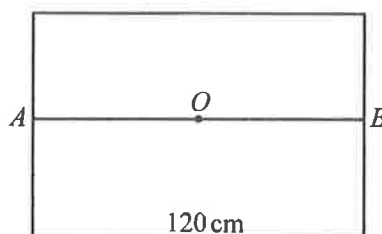
[N16/II/10]

6. A train travelling on a straight track passes a signal X with speed 30 m/s and, 20 seconds later, passes another signal Y with speed 10 m/s. During the journey from X to Y the acceleration, a m/s², of the train is given by $a = kt - 2$, where k is a constant and t seconds is the time after passing X .
- Show that $k = 0.1$. [5]
 - Find the distance between the signals X and Y . [4]

[N15/I/10]

Advanced

7.



The diagram shows a computer screen 120 cm wide. A dot oscillates in a straight line on the screen between the two points A and B on the edge of the screen. AB is parallel to the base of the screen. The displacement of the dot from O , the centre of AB , is modelled by the equation $x = a \sin nt$ where t is the time in seconds after passing through O and a and n are constants. The time for one oscillation is 6 seconds.

- Explain why $a = 60$ and show that $n = \frac{1}{3}\pi$. [3]
- Obtain an expression for the velocity of the dot at time t and hence deduce its maximum speed. [3]
- Find the magnitude of the acceleration of the dot when it is at A . [2]

[N22/I/12]

8. A girl cycles up a hill and passes a fixed point A . The distance, d m, that she cycles in t s after she passes A is given by

$$d = 840 \left(1 - e^{-\frac{t}{80}} \right) - 2t \quad \text{for } t \geq 0.$$

- Find the girl's speed and acceleration 10 seconds after she passes A . [6]
- Explain what the sign of the acceleration indicates. [1]

When she reaches point B her speed is 1.5 m/s and she gets off her bicycle. Given that A is 500 m from the top of the hill,

- calculate the distance she has to push her bicycle to reach the top of the hill. [4]

[N18/II/7]